# Survey on IOT Based Pothole Detection

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***Abstract*— Potholes can cause damage like flat tires and wheels damage, vehicle collisions, and severe accidents, this has become an intimidating problem in present days. On this subject, the project involves detecting potholes on roads and maintaining a database of the coordinates of that specific place. This is often achieved by designing a device embedded in the vehicle. When there's a pothole approaching, the device scans using its Infrared and Ultrasonic sensor, alerting the driver on time. This method becomes useful during the rainy season where the roads are crammed with rainwater and low visibility from fogging up during winter. Further, when there is confirmation of a pothole, introduce a mechanism that allows the detector to locate the position of the pothole via GPS (Global Positioning System) and coordinates of that place are noted and stored in a database. By using the system, the driver can make proper decisions and choose the best-suited route. Other beneficiaries include ambulances and local departments, as they address time-sensitive cases and transport of delicate instruments like medical equipment. The knowledge within the system is made available to the general public as well as the municipalities and road maintenance agencies. Thus, awareness of the position of the potholes will help drivers to avoid those roads and be more careful while driving on those roads.**

***Keywords—Potholes, IoT, Infrared and Ultrasonic Sensors, GPS (Global Positioning System).***

## I. INTRODUCTION

India, as the world's second most populated country in the world, is said to be one of the busiest in terms of road traffic. About 90% of the country's passenger traffic and about 65% of the goods were carried by the Indian road network, which spans over five million kilometers. Road safety has turned into a major factor of utmost importance for the country's citizens with the exponential increase in the number of cars and the inhumanely congested Indian roads. According to statistics provided by the Indian government, around 1.2 lakh cases of deaths were reported due to negligence related to road accidents in 2020, with about 328 people dying every day on an average. Statistics show that over 3,564 people were killed by potholes and speed breakers last year. A pothole is considered to be more lethal than terrorism. Potholes are a very ordinary thing in India and as a matter of fact, a person can see potholes within every one kilometer in India. Potholes are associated with large volumes of very turbulent water by many geologists. Around 3,500+ people (almost more than 30% of the total road accidents) die due to potholes every year. It becomes really difficult to identify the potholes on the roads during rainy times as they get filled with rainwater and hence it becomes difficult for us to recognize them. People who travel at night also face this difficulty while coming across a pothole due to the lack of street lights or any other light source, which in turn leads to accidents. Thus, these road accidents can be reduced by the effective detection of potholes and humps on road. If these potholes can be detected in real-time, then it will be of great help for both two-wheeler and four-wheeler vehicle users which will ensure a much safer journey, especially during monsoon season when the potholes are covered with rainwater. This paper proposes an IOT based pothole detection system that is designed to gather road images by using an optical device installed on a vehicle, assists in detecting a pothole, and then displays it with the help of the LED attached to the vehicle. It is designed in such a way that the vehicle rider can clearly see it. During the rainy season, this system is much useful. It can also monitor and transfer data to servers for the benefit of others. The data related to the potholes are collected by the optical device fitted into the vehicle and it then sends the data to the server with the help of information concerning GPS location. An app for pothole road maps can be created to share the information with other vehicles, in case they don't have any facility to accommodate a system for image detection in their vehicle. Pothole data such as the location and severity of the corresponding potholes can be sent to the road management center with the help of this system.

## II. LITERATURE SURVEY

Buza et al. [1] proposed a new unsupervised vision-based method that does not require any additional equipment, training, or filtering. For rough estimation and identification of potholes, their method employs image processing and spectral clustering. Image segmentation, shape extraction using spectral clustering, and identification and extraction are the three processes in the proposed method. The proposed

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method was implemented in MATLAB and tested on 50 pothole photos from the Google image collection. The accuracy of estimating the surface area of a pothole was around 81 percent. This method can be used to make a rough estimate for pavement repairs and rehabilitation.

Yu and Yu [3] proposed the development of a vibration-based system with data collecting software for the preliminary evaluation of road conditions. The vehicle is subjected to impacting forces due to pavement distress such as cracks and rutting. The pavement surface conditions can be evaluated using the recorded responses of the testing vehicle while driving on the pavement. Because of its small storage requirements, cost-effectiveness, and compliance with automatic real-time data processing, this system is attractive. However, it lacks the entire information of distress features as compared to the video-based approach.

Erikson et al [4] investigated a mobile sensing application for detecting and reporting road surface characteristics. The Pothole Patrol (P2) system was created to collect data from three-axis acceleration sensors and GPS devices installed in cars. P2 was installed in around 7 taxis in the Boston region. They used a basic machine-learning strategy to identify potholes and other significant road surface irregularities from accelerometer data. Detections were sent to central servers via opportunistic Wi-Fi connections provided by participating open Wi-Fi access points or, when available, a cellular data service.

Android smartphones with inbuilt accelerometers were used for real-time detection by Artis Mednis, et al [5] Accelerators are mainly based on the magnitude of vibrations. The Z-axis acceleration amplitude is measured using Z-thresh. Z-diff is used to obtain the difference between two amplitudes. Standard deviation is obtained using STDEV (Z) in the case of vertical axis acceleration. To identify the potholes accurately G-Zero is used.

Effective detection of potholes with the usage of a laser line striper and camera was suggested by Sandeep Venkatesh et al. [6]. A systematic database consisting of various pothole locations is created. To share data via messages to nearby vehicles, dedicated short-range communication is used.

Clubbing of smart transport system through Zigbee module along with pothole detection using ultrasonic sensors was suggested by Shambhu Hegde, et al [7]. But in this system, the quick alert system is not easy as there is some delay in its process.

Moazzam et al. [8] advised using the Kinect sensor because it is less expensive (because it takes less computing labor). The Kinect sensor includes an RGB camera and an IR camera for recording RGB images that may be processed systematically using MATLAB.

Pothole detection using laser technology was suggested by X. Yu et al [9]. Based on the abnormalities on road, images are captured and then they are detected with any laser source deformations.

Pothole detection using vision-based cameras was suggested by Sachin Bharadwaj et al. [10]. Only 2-D images of required potholes are captured by these cameras. The captured images are then analyzed with the help of MATLAB. This approach is restricted to proper lighting conditions.

## III. ANALYSIS OF EXISITNG METHODS

*A. IoT based Potholes and Drainages Detection System, developed by Divyani Dongre, Monali, Rima Kumbhalkar, Alisha Gosavi, and Prof. Ashish p Nanotkar Depar, Department of IT, JD College of Engineering and Management, Nagpur, DBATU*

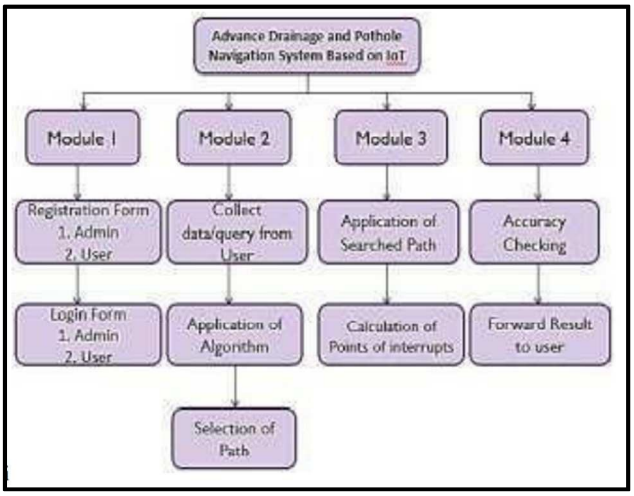



Fig. 1. Modules used in the System

TABLE I

| Advantages | Disadvantages |
|---|---|
| The potholes and drainage systems are easily detected. | Difficult to set the threshold value to detect potholes and drainage. |
| Can find out the current road conditions. | There may be network errors |
| The navigation system can be used to inform regarding traffic and accidents. | |
| It can be used to inform authorities to plan the engineering work of roads. | |

*B. IoT Based Real-Time Potholes Detection System Using Image Processing Techniques Kulwant Singh, Sristy Hazra, Chandra Mukherjee, Sushanth. G, Sanjith Gowda*

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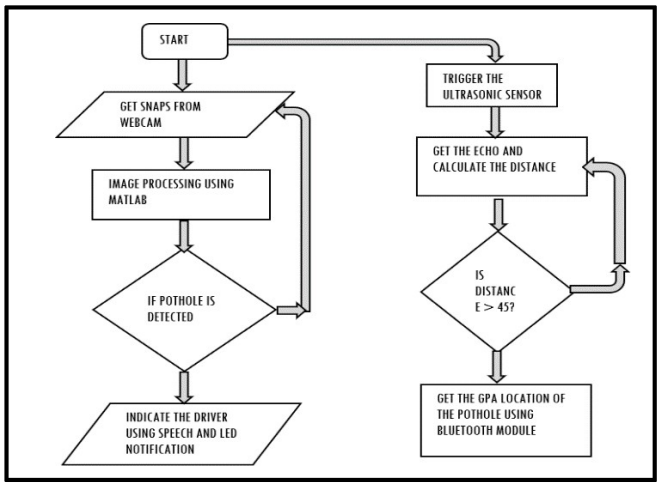



Fig. 2. Process Flow of the System

TABLE II

| Advantages | Disadvantages |
|---|---|
| The system effectively identifies humps and potholes helping drivers have a smooth ride. | Difficult to set the threshold value to detect potholes and drainage. |
| Notifications are sent to avoid accidents. | There may be network errors |
| The solution is efficient in detection in rainy seasons in low visibility. | |
| The information can help the maintenance of roads. | |

## IV. METHODOLOGY

Sensing, Communication, and Localization are the three subsystems that make up the proposed system. These subsystems are centered on data and are self-contained. Data is generated by the sensing system; data is collected, coordinated, and distributed by the communication subsystem; and finally, the data is used by the localization subsystem to provide information for the driver. Different methods can be used to realize and implement these subsystems. In this part, several of the methods are described, along with their benefits and drawbacks. In this section, we go over the subsystems and their benefits and drawbacks.

1. Data generation is the responsibility of the Sensing subsystem. In this instance, the data would be about the location of the pothole, the severity of the pothole, and so on. For this subsystem, two ways are being considered: one is based on vision, while the other is based on vibration.

### *A. Vision-Based Method:*

The camera is utilized as a sensor in this technique to scan the road for potholes. The camera captures photos in real-time. Image processing methods like edge detection, which demand a lot of processing time and power, are applied to the images collected by the camera. There are a variety of design options. When compared to software-based methods, the use of special Digital Signal Processors or Application Specific Integrated Circuits improves performance. However, due to limitations such as windowing convolution for the image processing algorithm, its response time is still high. This approach has an advantage over others since it can detect or sense potholes without actually passing through them. Pothole characterization can be done based on the magnitude of the pothole. In terms of vision-based methods, we have two options:

a. 2-D Image-based approach
b. Video-based approach

The following is the architectural decision for the Sensing Subsystem:
Based on the comparison in the table below, the vibration-based technique is preferable due to the device's relatively short processing time, response time, and cost of maintenance.

TABLE III

| | VIDEO-BASED | VIBRATION-BASED |
|---|---|---|
| SENSOR | Camera | Accelerometer |
| RESPONSE TIME | High | Low |
| SENSING PERIOD | While approaching the pothole | While going through the pothole |
| PROCESSING | Complex image processing algorithms | Readings are directly used |
| MAINTENANCE COST | High | Low |
| CHARACTERIZATION OF POTHOLE | Based on size | Based on vibrations |

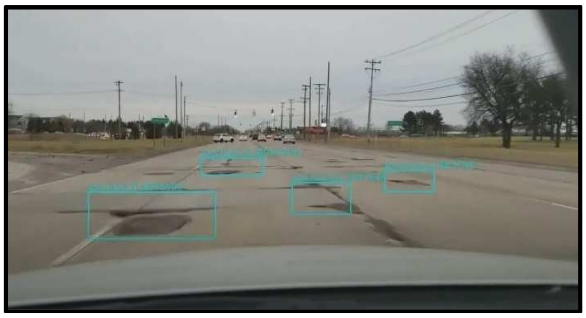


Fig.3: Vision-Based Pothole detection

### *B. Vibration-Based Method:*

To detect potholes, this system uses the 'Accelerometer' technique. The term "accelerometer" refers to a device that measures the total specific external force acting on a sensor. If we use a stationary apparatus, for example, it will display a reading that corresponds to the gravitational pull of the earth. For an accelerometer falling freely in the vacuum, it will show a zero value. The instrument is widely used to determine tilt or vibration. It is mostly used to detect inclination and modify the display view in some cameras and the iTouch. Potholes can be identified by the amount of variation in accelerometer readings.

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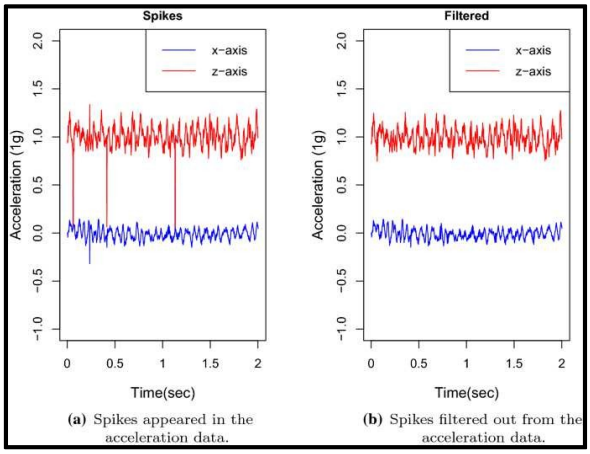


Fig.4: Vibration-Based Pothole detection

### C. *Ultrasonic Based Method*

Ultrasonic sensors are used to measure the characteristics of sound waves that have a frequency that is higher than what the human ear can hear. The receiver, ultrasonic transmitter, and control circuit are all part of the HC-SR04 module. This instrument is used to determine the distance between two entities. The module automatically sends a 40 kHz square wave and detects the return pulse signal. The distance is estimated using the time it takes for the sent signal to return to the receiving end.

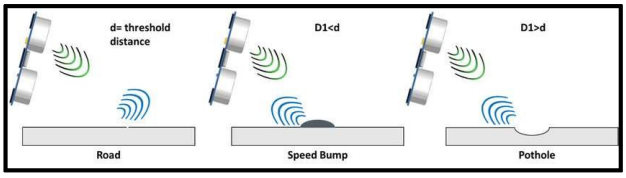


Fig.5: Ultrasonic Based Pothole detection

1) Communication Subsystem:
Based on the factors that drive communication, the table below compares three techniques. Because of its coverage and flexibility, it is recommended that the communication subsystem employ the Hotspot technique.

TABLE IV

| | ENTIRE Wi-Fi COVERAGE | PUBLIC TRANSPORT AS AP | HOTSPOT |
|---|---|---|---|
| DEPLOYMENT | Deploy AP wherever coverage is required. | Only on public transport vehicles. | A particular location where vehicle density is high. |
| COVERAGE | Wherever AP's are deployed. | Wherever public transport vehicles commute. | Wherever the vehicles can go. |
| COST OF AP's DEPLOYMENT | High | Medium | Low |

Feedback Subsystem, which is a part of the Communication Subsystem, has the following architectural choice:

The Feedback Subsystem guarantees that the information with the access point is updated in a timely and correct manner. The following are a few of the architectural options for sensor deployment:

i) In each vehicle: Each vehicle is equipped with a sensor. However, this causes the channel's load to increase. This option will only be viable in dynamic settings such as congestion avoidance, where every vehicle's movement is relevant. It has a significant benefit in collision detection systems where vehicles can make their own decisions.

ii) Public transportation: By proposing the deployment of sensors in public vehicles such as taxis and buses, the method reduces duplication. This architecture can be utilized for somewhat dynamic settings such as the distribution of appropriate congestion information, pothole detection, and so on.

iii) Pilot vehicle: A pilot vehicle is a vehicle designed for a specific function. It is expected to travel through the roads regularly, collecting data about potholes. This design is beneficial in less dynamic situations, such as when data collecting from roadside Apps is required. Pilot vehicles can be utilized when there is a non-zero chance that road conditions will change dramatically owing to heavy rains, maintenance work, earthquakes, road construction, and so on.

2) Subsystem for Localization
The localization subsystem uses the data provided by Access Point to determine the location of the pothole and inform the driver about it. Localization can be difficult when the Access Points are out of range. In such instances, the car must either figure out how to navigate on its own or receive external assistance to locate itself. The Global Positioning System (GPS) is one of the most common and effective tracking technologies available (GPS). Vehicles can quickly determine the position of potholes on the GPS map using data obtained from the access point.

## V. PROPOSED METHODOLOGY

Implementation of potholes detection is divided into two aspects: Potholes Detection using an IOT System and a Mobile Application that alarms the rider/driver of potholes onreaching coordinates stored in the database by the detection system.

### A. *Potholes Detection using IoT System*

The hardware components used are:

- GPS Module
- Zigbee Transmitter and Receiver
- Accelerometer Sensor

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- Vibration Sensor
- ARM 7 Development Board

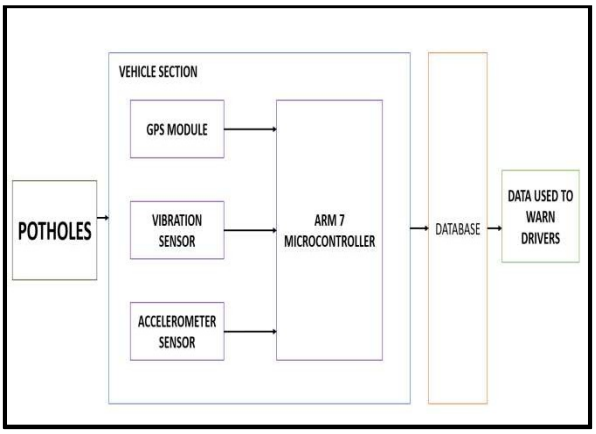



Fig. 6. Connection of Hardware Components

Fig. 1 indicates the sensors embedded in the vehicle. On detection of a pothole, the values from the vibration and accelerometer sensors are considered. A threshold value is set for both sensors. A pothole is defined as a value that exceeds the threshold value. The vehicle's direction is detected by the accelerometer sensor. X-value indicates right and left direction, y-value indicates forward and backward, and z-value indicates the movement of a vehicle in a pothole. If the values of both the sensors exceed the threshold values, the GPS captures the longitude and latitude values and stores them in the database. The database comprises of columns Pothole depth, Longitude, and Latitude coordinates. The coordinates are also sent to the Municipality to intimate them to repair the same. The database is connected to the mobile application developed.

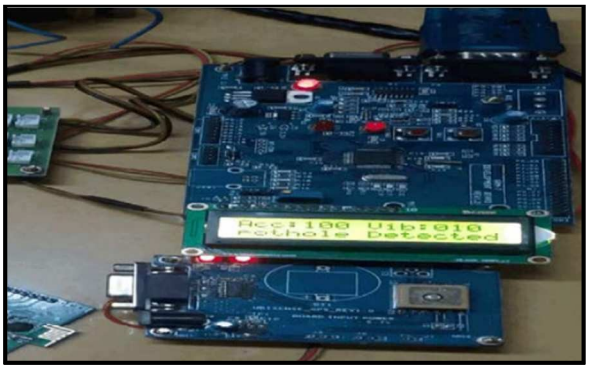



Fig. 7. IoT System for Pothole Detection

### B. *Mobile Application Development*

Android Studio is used to develop an android application that notifies the driver of potholes. The application is programmed to access the current longitude and latitude coordinates. The distance between the present longitude and latitude coordinates and the longitude and latitude coordinates of the potholes contained in the database is calculated using a distance algorithm. If the algorithm's determined distance is greater than the threshold value, the driver is alerted to the potholes via audio and notification.

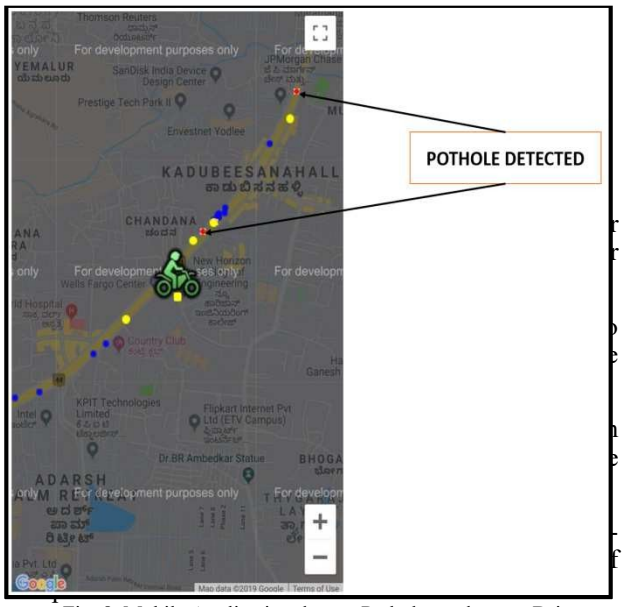



Fig. 8. Mobile Application detects Potholes and warns Driver

- The warning audio and notifications sent to the driver via the application help the driver avoid potholes or slowdown on crossing the potholes.
- The data collected by the IoT System can be sent to the appropriate authorities, who can then take the appropriate actions.
- The system helps drivers drive passing potholes in rainy and foggy conditions easily with the help of the mobile application alert system.
- Setting the threshold had to be done using the trial-and-error method considering the varying depths of potholes.
- There may be some ambiguity between potholes, cracks, and humps.

| POTHOLE | LATITUDE | LONGITUDE |
|---|---|---|
| 4 CM | 12.9401° N | 77.5812° E |
| 6 CM | 12.5653° N | 77.3397° E |
| 6.2 CM | 12.5872° N | 77.3491° E |
| 3.1 CM | 12.5238° N | 77.6353° E |
| 2 CM | 12.5606° N | 77.3945° E |
| 2.3 CM | 12.5843° N | 77.9721° E |

Fig. 9. Data Collected from the IoT System

## VI. CONCLUSION AND FUTURE ENHANCEMENTS

### A. *Conclusion*

- The proposed system in this study is a smartphone application for detecting and reporting road surface conditions.
- It serves by identifying the potholes on the roads and sends alerting signals to the driver, thus helping avoid accidents, and also stores the location of potholes on the database.
- Each road vehicle can use data to access informationand estimate speed based on potholes and humps on the road.

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- It sends a voice-based alarm message and notification to the motorist, encouraging them to be even more cautious. This technique is particularly useful during the rainy season when roads are flooded with rainwater because alerts are delivered by retrieving data from a database.
- Also useful during winter from low visibility. Hence, it avoids terrible accidents due to bad roads conditions.
- Mapping tools such as Google Maps or Open-Street-Map could connect this database to a network infrastructure. It can be utilized by government agencies to keep these roads in good repair.

### B. *Future Enhancements*

- An API can be developed by which the government can access the database.
- The distinction between bumps and potholes can be improved.
- Notification regarding the severity of potholes can also be provided to help drivers make more informed choices.
- Reduce misclassifications.
- Adapting ML techniques to classify data that can help to adapt the nature of road and vehicle type.

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